

SQL Skill Assessment

Use-case based questions covering core SQL concepts

8 Questions | 60 Minutes | 5 Sample Tables

1 | Sample Database Tables

All 10 questions are based on the five tables below. Study the schema and sample data carefully before attempting any question.

Table 1 — employees

emp_id	name	dept_id	salary	manager_id	hire_date
1	Alice	10	90000	NULL	2018-03-15
2	Bob	20	75000	1	2019-07-01
3	Carol	10	85000	1	2020-01-10
4	David	30	60000	2	2021-05-20
5	Eve	20	70000	2	2019-11-30
6	Frank	30	65000	3	2022-08-05
7	Grace	10	95000	1	2017-06-22
8	Hank	NULL	55000	3	2023-01-18

Table 2 — departments

dept_id	dept_name	location
10	Engineering	New York
20	Marketing	Chicago
30	Finance	Dallas
40	HR	Seattle

Table 3 — orders

order_id	customer_id	order_date	amount	status
101	C01	2024-01-05	1200.00	Completed
102	C02	2024-01-10	450.00	Completed
103	C01	2024-02-15	800.00	Pending

104	C03	2024-02-20	2300.00	Completed
105	C02	2024-03-01	150.00	Cancelled
106	C03	2024-03-10	900.00	Completed
107	C04	2024-03-15	3100.00	Pending
108	C01	2024-04-02	670.00	Completed

Table 4 — products

product_id	product_name	category	price	stock
P01	Laptop	Electronics	1200.00	30
P02	Desk Chair	Furniture	350.00	50
P03	Monitor	Electronics	450.00	25
P04	Keyboard	Electronics	80.00	100
P05	Bookshelf	Furniture	200.00	40
P06	Headphones	Electronics	150.00	0
P07	Desk Lamp	Furniture	60.00	0

Table 5 — sales

sale_id	emp_id	product_id	sale_date	qty	sale_amount
S01	2	P01	2024-01-08	2	2400.00
S02	5	P03	2024-01-15	1	450.00
S03	2	P04	2024-02-01	5	400.00
S04	4	P02	2024-02-10	3	1050.00
S05	5	P01	2024-02-20	1	1200.00
S06	4	P05	2024-03-05	2	400.00
S07	6	P03	2024-03-12	2	900.00
S08	2	P06	2024-03-18	1	150.00
S09	6	P04	2024-04-01	3	240.00
S10	5	P02	2024-04-10	1	350.00

2 | Questions

Write SQL queries to answer each question. The topic tag indicates the concept being tested.

Q1 | Employee department listing

List the names of all employees along with their department name and location. Include employees who have not been assigned to any department.

► Expected Output

name	dept_name	location
Alice	Engineering	New York
Carol	Engineering	New York
Grace	Engineering	New York
Bob	Marketing	Chicago
Eve	Marketing	Chicago
David	Finance	Dallas
Frank	Finance	Dallas
Hank	NULL	NULL

Q2 | Above-average earners

Find the names and salaries of all employees who earn more than the average salary of the entire company.

► Expected Output

name	salary
Alice	90000
Carol	85000
Grace	95000

Q3 | Outearning the Finance team

Retrieve the names of employees whose salary is greater than the maximum salary of every individual employee in the Finance department (dept_id = 30).

► Expected Output

name	salary
Grace	95000
Alice	90000

Carol	85000
Bob	75000
Eve	70000

Q4

For each employee, find their name, salary, and the average salary of their own department. Display only employees who earn above their department's average salary using subquery.

► Expected Output

name	salary	dept_avg_salary
Grace	95000	90000.00
Bob	75000	72500.00
Frank	65000	62500.00

Q5

Sales ranking per employee

For each sale, display the employee name, product name, sale amount, and the rank of that sale within each employee's sales ordered by sale amount descending. Also show the total sales amount per employee using a window function.

► Expected Output

name	product_name	sale_amount	rank_in_emp	total_emp_sales
Bob	Laptop	2400.00	1	2950.00
Bob	Keyboard	400.00	2	2950.00
Bob	Headphones	150.00	3	2950.00
David	Desk Chair	1050.00	1	1450.00
David	Bookshelf	400.00	2	1450.00
Eve	Laptop	1200.00	1	2000.00
Eve	Monitor	450.00	2	2000.00
Eve	Desk Chair	350.00	3	2000.00
Frank	Monitor	900.00	1	1140.00
Frank	Keyboard	240.00	2	1140.00

Q6

For each completed order, display the order ID, customer ID, order date, amount, and a 3-order rolling average of the amount (current row and the 2 preceding rows) ordered by order date. Round the rolling average to 2 decimal places.

► Expected Output

order_id	customer_id	order_date	amount	rolling_avg_3
101	C01	2024-01-05	1200.00	1200.00
102	C02	2024-01-10	450.00	825.00
104	C03	2024-02-20	2300.00	1316.67
106	C03	2024-03-10	900.00	1216.67
108	C01	2024-04-02	670.00	1290.00

Q7

Products that have been sold

Find the names and categories of all products that have been sold at least once. Use the EXISTS keyword.

► Expected Output

product_name	category
Laptop	Electronics
Desk Chair	Furniture
Monitor	Electronics
Keyboard	Electronics
Bookshelf	Furniture
Headphones	Electronics

Q8

GROUP BY + HAVING — COMMON INTERVIEW
Departments with more than 2 employees

Find each department that has more than 2 employees, along with the number of employees and the highest salary in that department.

► Expected Output

dept_name	employee_count	max_salary
Engineering	3	95000

Q9 Explain Normalization

Q10 Explain CTE Vs Views